

Program 3:

Title: Devise a mobile application to set alarm clock.

Problem Description: The problem involves alerting the user in the form of notification /sound/ message once the alarm goes off. The user sets the alarm using proper UI and close the app.

Method: The program involves use of **Broadcast receiver** to receive the broadcasts from android system for the alarm. **Alarm Manager API** to set the alarm. **Pending Intent** for communication between the receiver and the activity for alert. UI is designed using TimePicker dialog.

Solution

Work flow of the Program

1. User clicks **Set Alarm button**.
2. **TimePicker dialog** appears.
3. User selects **alarm time**.
4. **AlarmManager schedules the alarm**.
5. At the specified time, **BroadcastReceiver is triggered**.
6. User receives **notification / message**.

Method

The program uses:

1. **Broadcast Receiver** to receive the alarm event.
2. **AlarmManager API** to schedule the alarm.
3. **PendingIntent** for communication between activity and receiver.
4. **TimePicker Dialog** for selecting alarm time.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/setAlarm"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set Alarm"/>

</LinearLayout>
```

MainActivity.kt

```
package com.example.alarmapp

import android.app.*
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.TimePicker
import java.util.*

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val button = findViewById<Button>(R.id.setAlarm)

        button.setOnClickListener {

            val timePicker = TimePickerDialog(this,
                { _, hour, minute ->

                    val calendar = Calendar.getInstance()
                    calendar.set(Calendar.HOUR_OF_DAY, hour)
                    calendar.set(Calendar.MINUTE, minute)

                    val intent = Intent(this, AlarmReceiver::class.java)
                    val pendingIntent = PendingIntent.getBroadcast(
                        this, 0, intent, PendingIntent.FLAG_IMMUTABLE)

                    val alarmManager = getSystemService(ALARM_SERVICE) as AlarmManager

                    alarmManager.set(
                        AlarmManager.RTC_WAKEUP,
                        calendar.timeInMillis,
                        pendingIntent
                    )

                }, 12, 0, true)

            timePicker.show()
        }
    }
}
```

AlarmReceiver.kt

```
package com.example.alarmapp

import android.content.BroadcastReceiver
import android.content.Context
import android.content.Intent
import android.widget.Toast

class AlarmReceiver : BroadcastReceiver() {

    override fun onReceive(context: Context, intent: Intent) {

        Toast.makeText(context, "Alarm Time Reached!", Toast.LENGTH_LONG).show()

    }
}
```

AndroidManifest.xml

```
<receiver android:name=".AlarmReceiver"/>
```

Output

1. User clicks **Set Alarm button**.
2. **TimePicker dialog** appears.
3. User selects the time.
4. When the time is reached, the **Broadcast Receiver triggers and displays a message:**
“Alarm Time Reached!”